EE6320 RF Integrated Circuits
Project: VCO Design

VCO Performance Summary Table

Design Parameter	Design Metric	Performance	Specification
Output Amplitude (Vpp/2)	freq = 6900 MHz	1.4816V	≥ 1.2 V
	freq = 7000 MHz	1.6788V	≥ 1.2 V
	freq = 7100 MHz	1.71V	≥ 1.2 V
Phase Noise (1 MHz Offset)	freq = 3450 MHz	-126.81 dBc /Hz	≤ -122 dBc/Hz
	freq = 3500 MHz	-128.14 dBc /Hz	≤ -122 dBc/Hz
	freq = 3550 MHz	-128.09 dBc /Hz	≤ -122 dBc/Hz
Phase Noise (20 MHz Offset)	freq = 3450 MHz	-153.40 dBc /Hz	≤ -154 dBc/Hz
	freq = 3500 MHz	-154.054 dBc /Hz	≤ -154 dBc/Hz
	freq = 3550 MHz	-154.059 dBc /Hz	≤ -154 dBc/Hz
Tuning Range	Total Tuning Range	248 MHz	≥ 200 MHz
	Number of bits in coarse tuning	3 bits	-
	Voltage range in fine tuning	0 - 1.2	-
	Average KVCO	68 MHz	70 MHz/V
	% variation in KVCO	14.55 %	minimal
Power Consumption (7000 MHz)	VCO average power consumption (excluding bias)	1.2 * 8m = 9.6 mW	minimal
	Bias circuit	Not applicable	minimal
Other	Sum of all capacitances (in capacitor bank)	22f * (1+2+4) * 2 = 308cf	-
	Net inductance used	Nothing other than tank inductance	-
	Simulator used	ADE L	-

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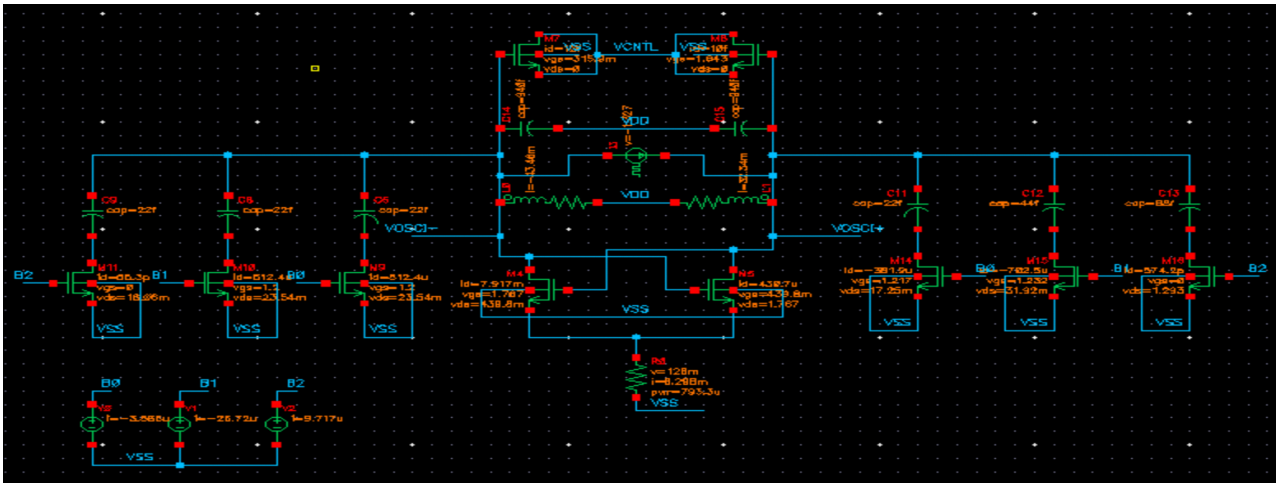
Component Values Table

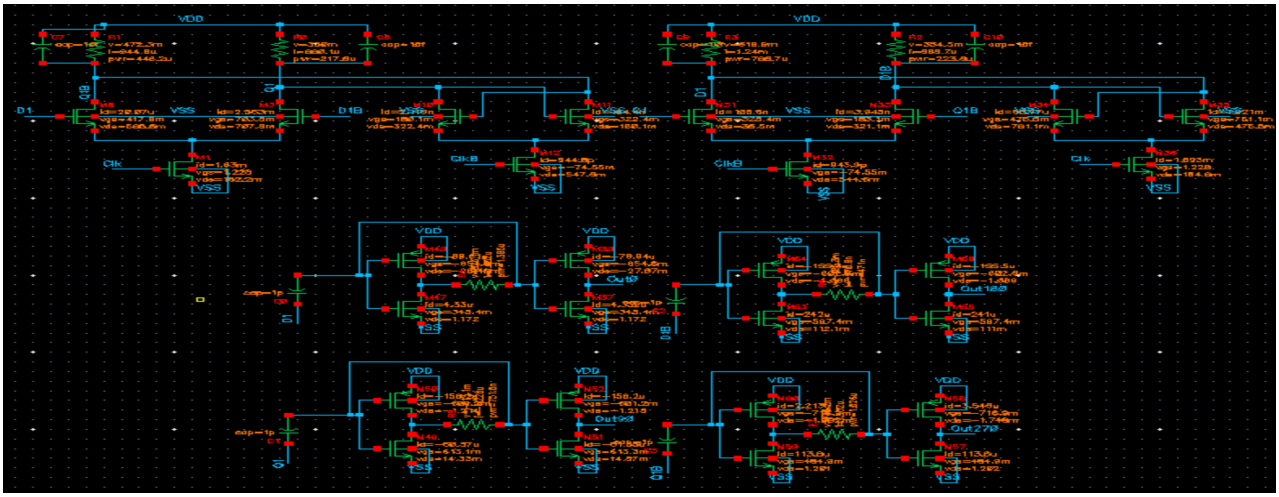
Component	Value <report only those which are used by you>
(W, L) Current Mirror MOS	**Not used**
(W, L) Tail MOS	**Not used**
(R, C) Low Pass Noise Cancelling Filter	**Not used**
Tail Capacitance	**Not used**
(W, L) Cross Coupled MOS(s)	35 uM and 480 nM
(W, L) Switching MOS(s)	15u , 120n
(C1, C2, C3) Capacitor Bank	22f , 44f, 88f
(W, L) Varactor MOS	25uM, 120nM
(R, L, C fixed) Drain Tank For 7Ghz , Q = 20	330p * 2 = 660 pH and 940f /2 = 470 fF R = 580.567 ohms
Tail resistor	20 ohm
(V bias, I bias)	**Not applicable**

Divide by Two circuit Component values

Component	Value <report only those which are used by you>
(W, L) Tail MOS	20u , 120n
(R, C) Drain Filter	500 ohm, 10 fF
(W, L) Cross Coupled MOS(s)	15u , 120n
(W,L)Inverter PMOS	10u , 120n
(W,L)Inverter NMOS	5u , 120n
(R, C) Biasing Inverter	10k ohm , 1pf

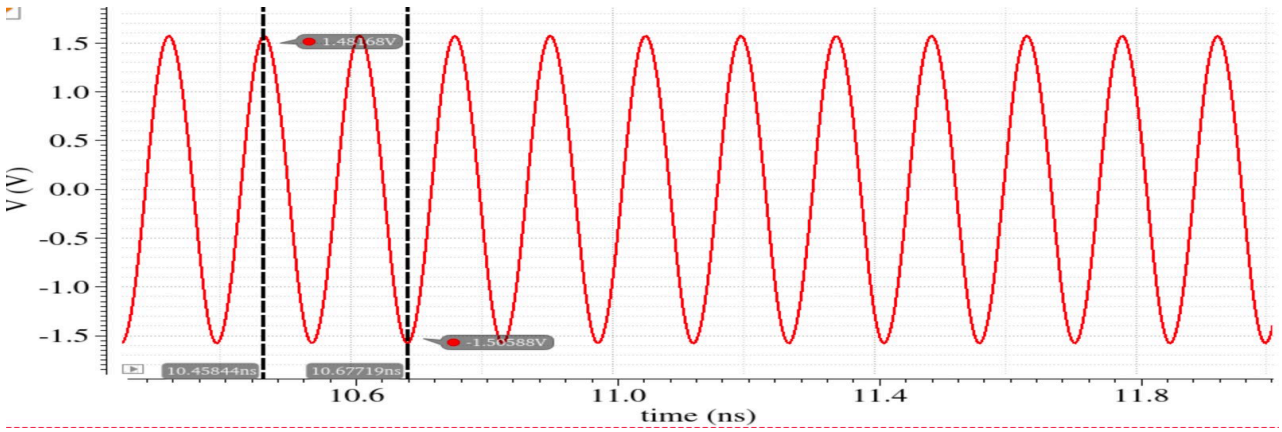
Schematic



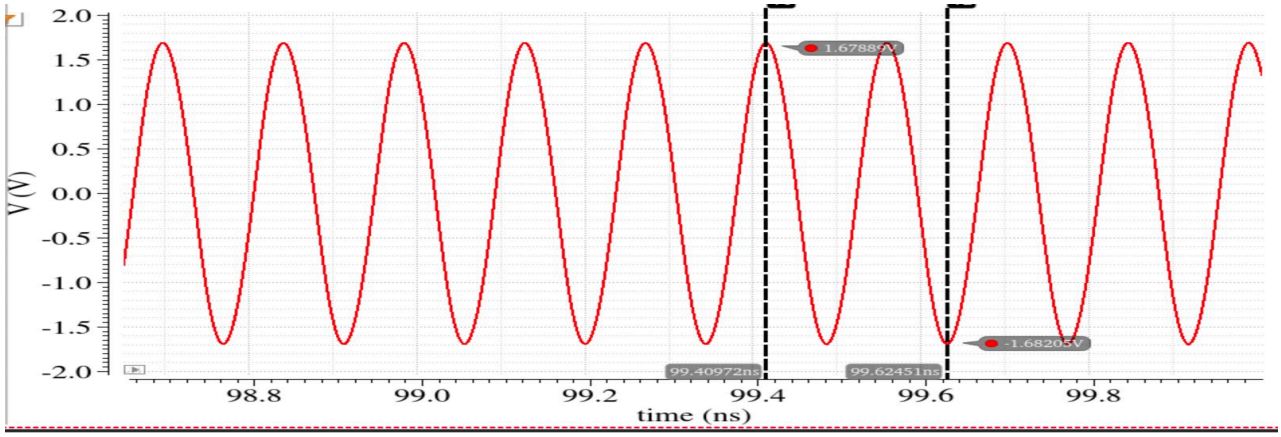


Output Amplitude Plots
(differential settled waveforms)

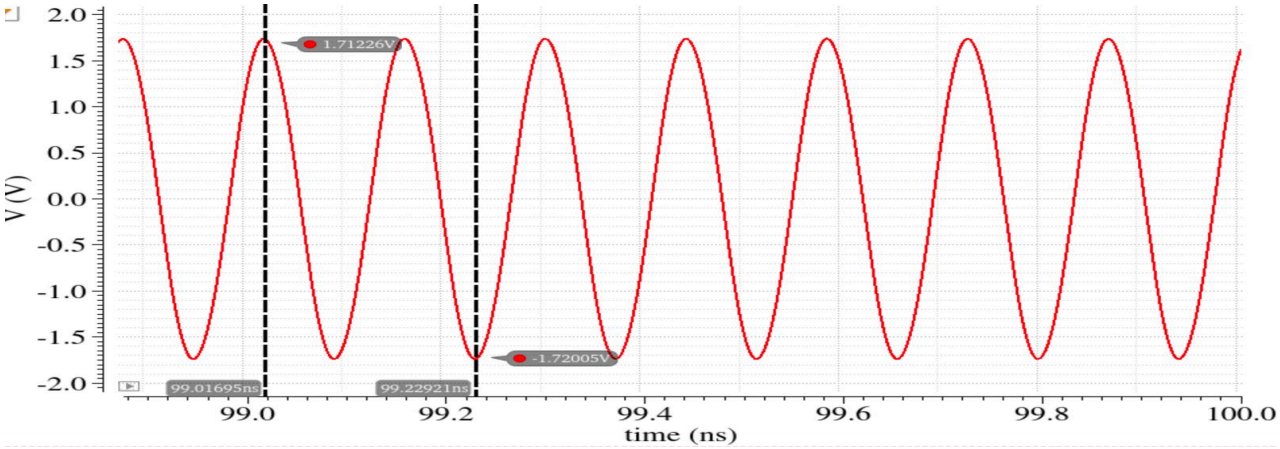
At 6900 MHz



At 7000 MHz

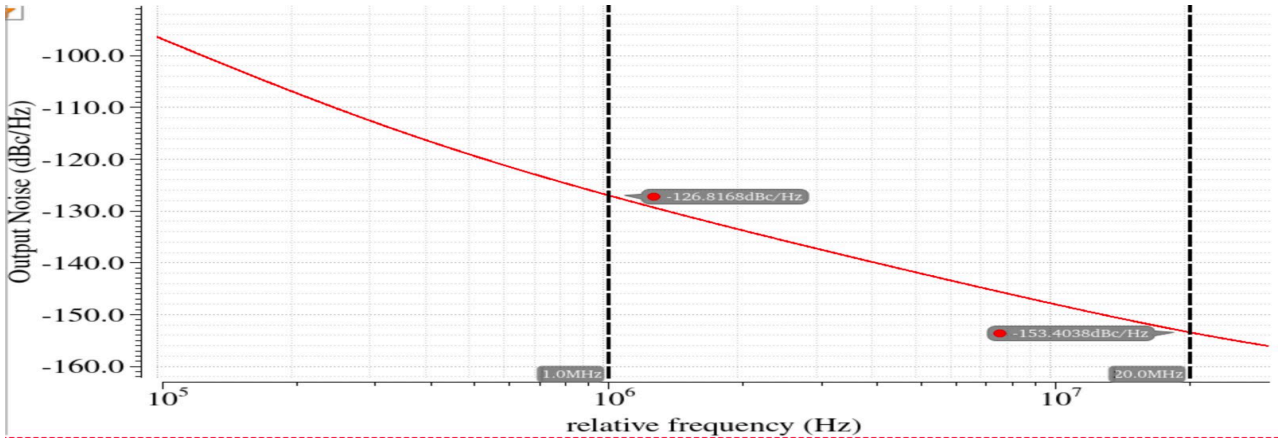


At 7100 MHz

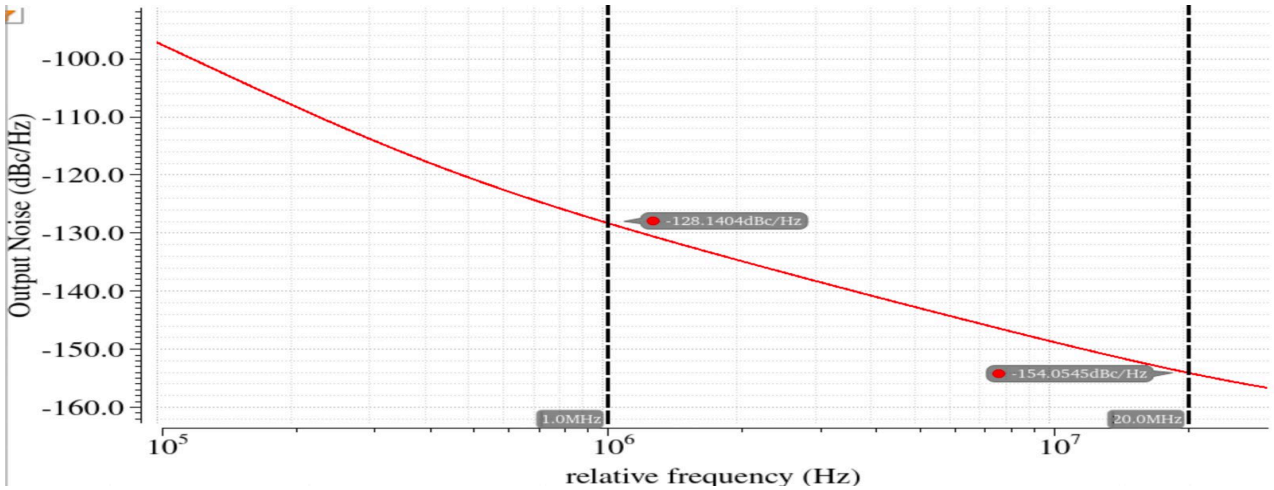


Phase Noise Plots
(at 1 MHz and 20 MHz)

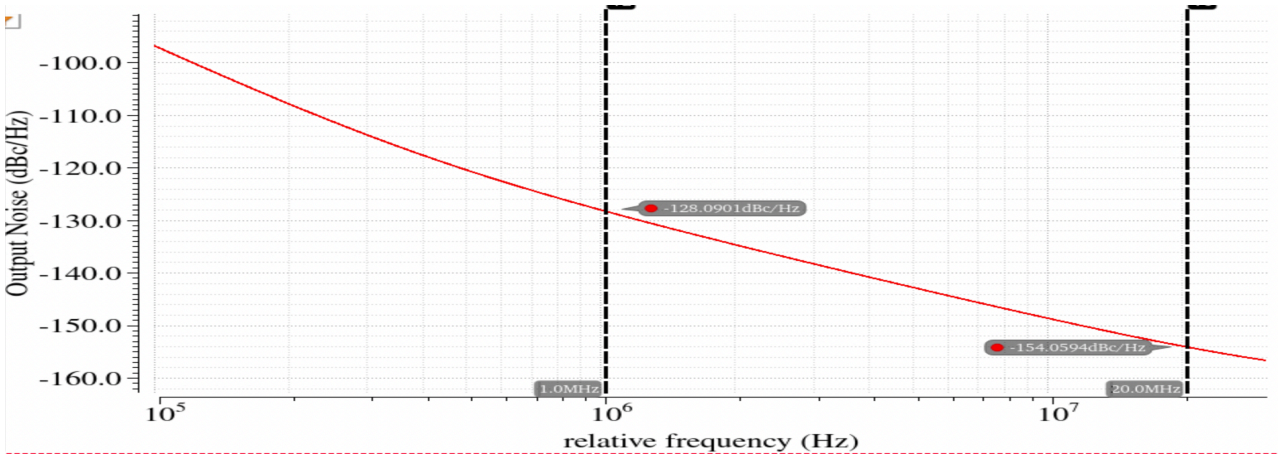
At 3450 MHz



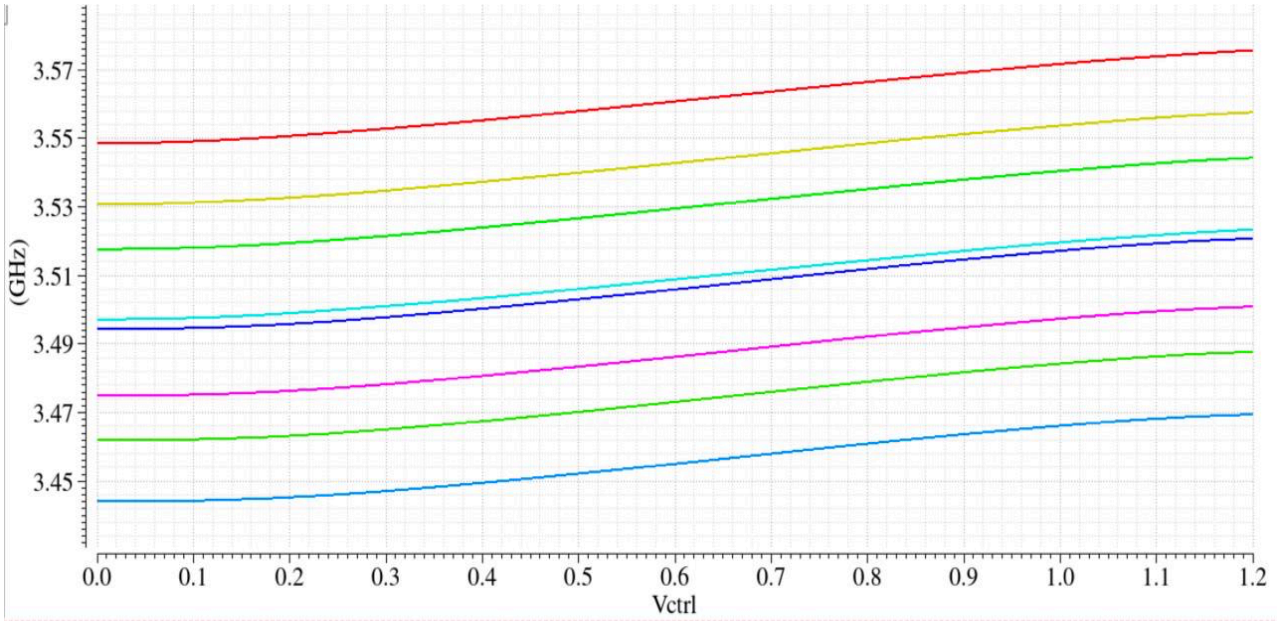
At 3500 MHz



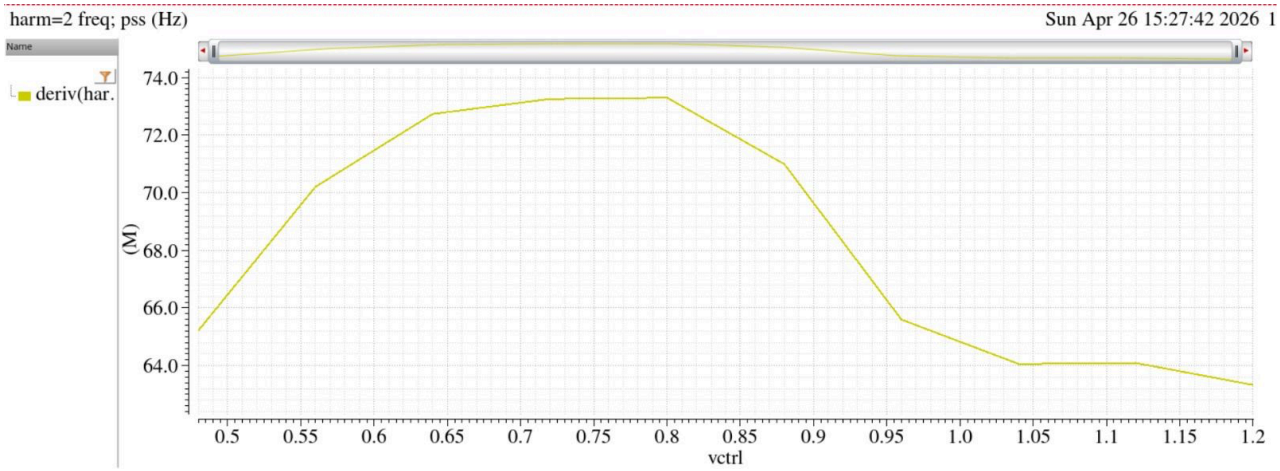
At 3550 MHz



Frequency Tuning Range Plot

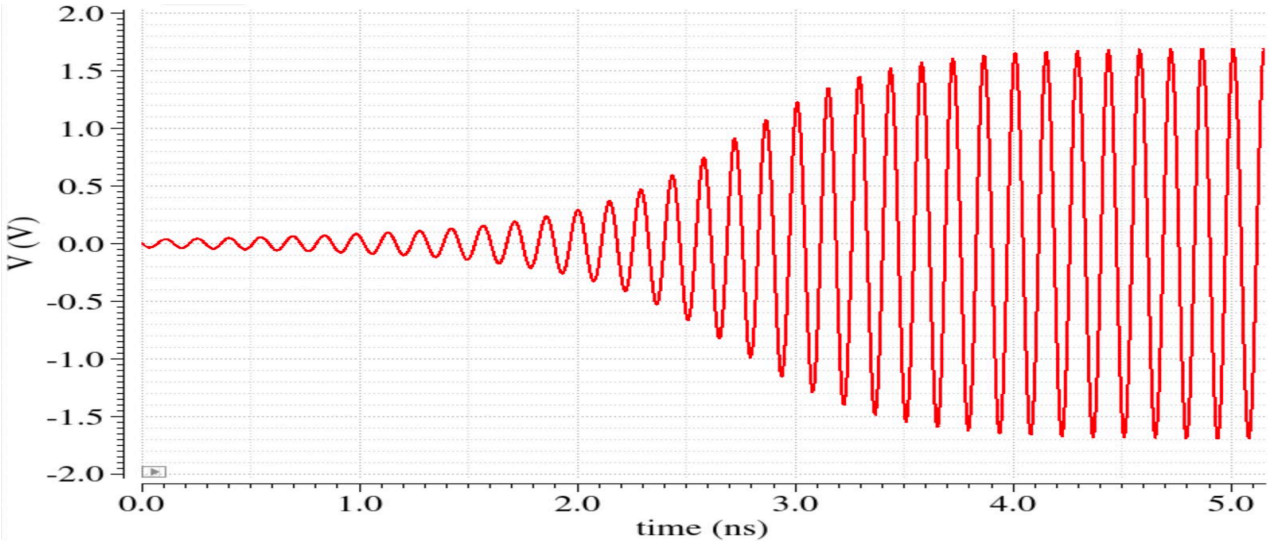


KVCO Plot



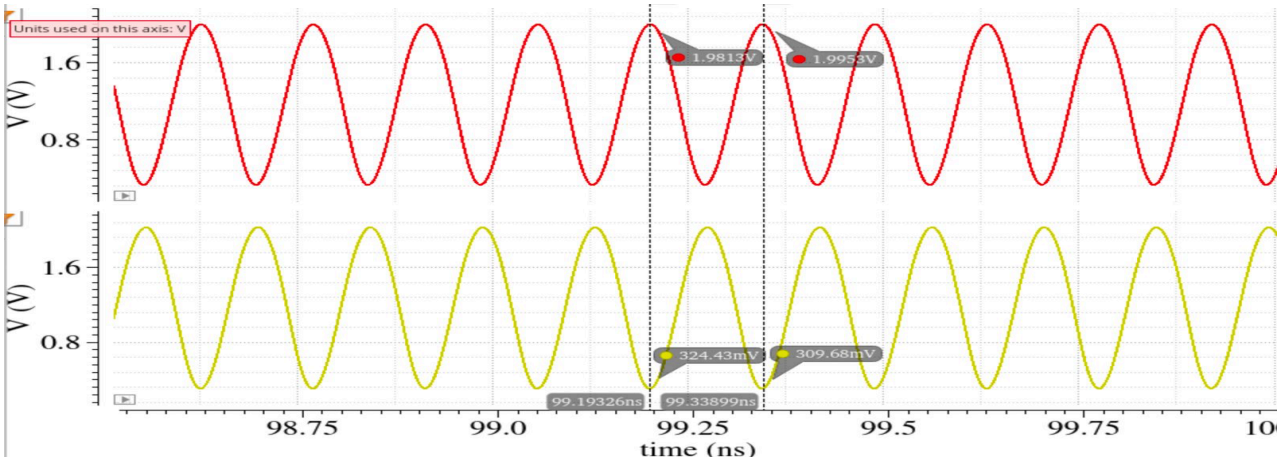
Initial Transient Response Plot

(for Vop at 7000 MHz)

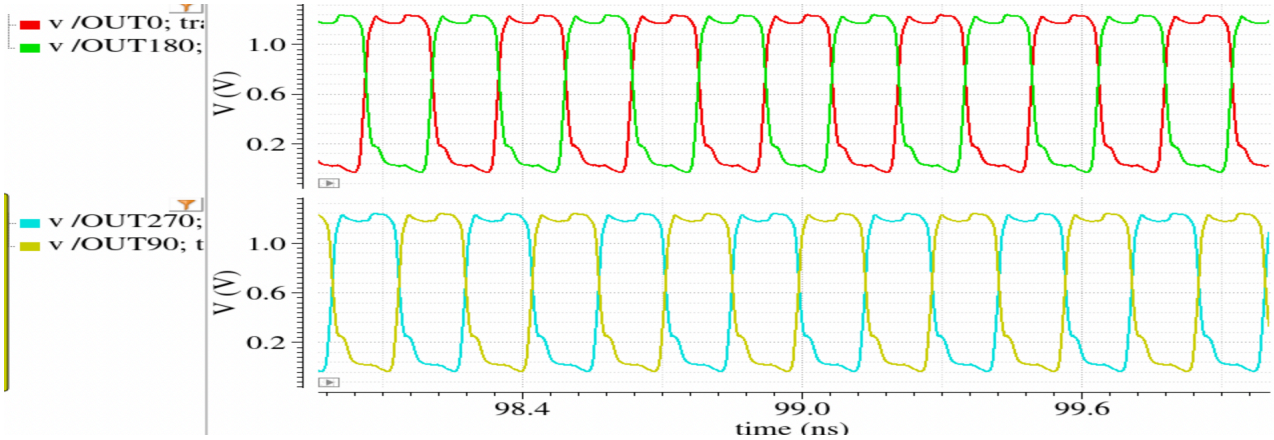


Single Ended Output Plot (Vop & Von)

(at7000 MHz)



Plots for 0, 90, 180 and 270 degree signals of Frequency Divider output



Design Procedure

Started with gmR>>1 for starting oscillations, started with gmR=3 and after it started oscillating, observed that phase noise is high because of tail current source mosfet and mirror mosfet flicker noise.
So I removed the current source mosfet and replaced it with a resistor and observed a good decrease in phase noise in the 1Mhz offset.
For the 20Mhz offset, I started increasing current and increasing sizes of transistors for the divider circuit as their thermal noise was getting added. Used Current mode latch for divider circuit followed by self biased inverters for quadrature output, which uses nearly **3mA** other than VCO consumption.

Path to Project Files

/home/ee23b114/cadence/EE6320_2026/ee6320_2026/VCO_TB

Note: All the plots/schematic shown above are for representational purposes only. Please replace them with the respective plots/schematic as obtained/designed by you.